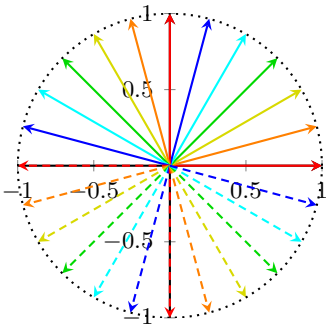
Exercise svdRose

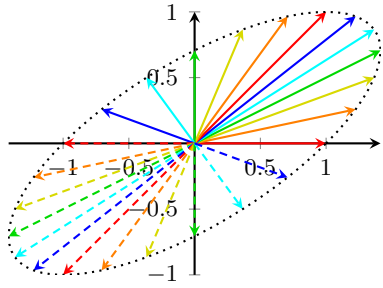
A. J. Roberts, July 21, 2025

Each plot below shows (unit) direction vectors $\boldsymbol{v}$ (left), and the corresponding vectors $A\boldsymbol{v}$ (right), for some specific matrix A. Estimate which *pair of orthogonal* directions $\boldsymbol{v}_1, \boldsymbol{v}_2$ give a *pair of orthogonal* result vectors $A\boldsymbol{v}_1, A\boldsymbol{v}_2$. By imagining this resultant pair as $\sigma_1\boldsymbol{u}_1, \sigma_2\boldsymbol{u}_2$ for orthogonal unit vectors $\boldsymbol{u}_1, \boldsymbol{u}_2$, estimate the singular values σ_1, σ_2 of the matrix A.

1, 0, 0, 1, 1

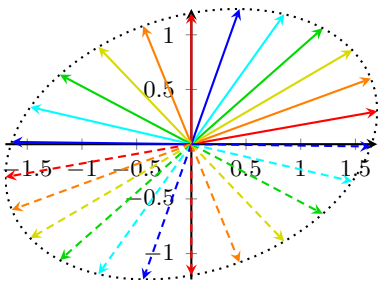


1, 1, 0, 1, 1

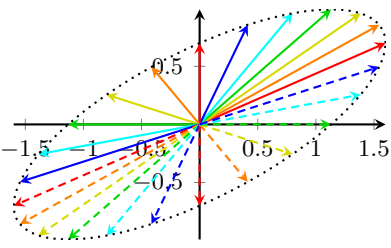


Answer cyan $\boldsymbol{v}_1 \approx \pm(0.5, 0.8)$ and $\boldsymbol{v}_2 \approx \pm(-0.8, 0.5)$, map to cyan $A\boldsymbol{v}_1 \approx \pm(1.4, 0.9) \approx \pm(0.8, 0.5)\cdot 1.6$ and $A\boldsymbol{v}_2 \approx \pm(-0.3, 0.5) \approx \pm(-0.5, 0.8)\cdot 0.6$. Thus singular values of A are $\sigma_1 \approx 1.6$ and $\sigma_2 \approx 0.6$.

1.7, 0, 0.3, 1.2, 2.04



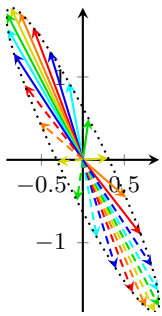
1.6, 0, 0.7, 0.7, 1.12

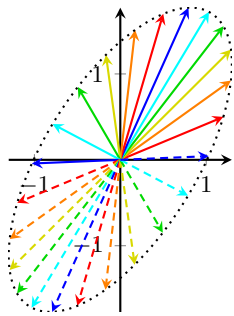


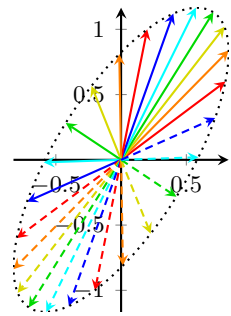
0.7, −0.6, −0.9, 1.6, 0.58

1.2, 0.5, 0.5, 1.7, 1.79

0.8, 0.2, 0.6, 1, 0.68







0.5, 0, 1, 1.9, 0.95

